

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 38)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

A low-lying coastal town in eastern England is at risk of flooding from rising sea level and more intense rainfall.

0 1

1 Describe **three** ways the town could adapt to the increased flood risk.

[3 marks]

1

2

3

0 1

2

Explain the circumstances in which managed retreat may be chosen instead of coastal defences.

[2 marks]

0 1

3

Describe **two** impacts of higher temperatures on transport infrastructure.

[2 marks]

1

2

0 1

4

Explain how carbon capture and storage (CCS) could reduce CO₂ emissions from a fossil fuel power station, and give **one** limitation.

[3 marks]

10

0 2 . 1

Define the term **persistence** as applied to a pollutant.

[1 mark]

0 2 . 2

Explain how the persistence of a pesticide is affected by **two** named environmental factors.

[4 marks]

5

0 3 . 1

Describe **three** effects of noise on human health other than hearing damage.

[3 marks]

1

2

3

03 . 2

Explain how noise can damage non-living objects.

[3 marks]

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03 . 3

Explain how noise can reduce the breeding success of birds.

[2 marks]

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03 . 4

Explain why military sonar is a threat to cetaceans.

[2 marks]

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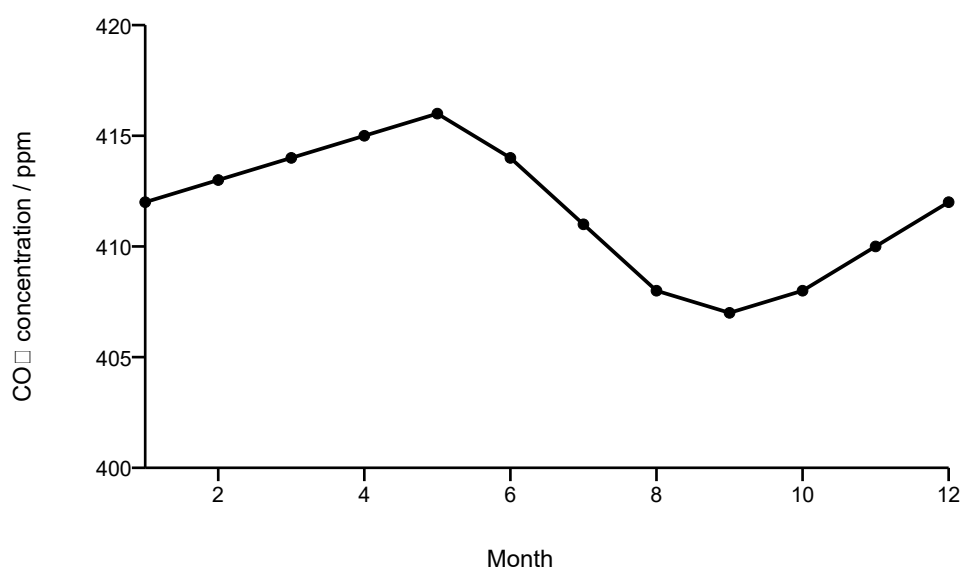
10

0 4

Table 1 shows the composition of the dry atmosphere. Figure 1 shows the changes in carbon dioxide concentration measured at a site in the northern hemisphere over one year.

Table 1

Gas	Percentage by volume / %
Nitrogen	78
Oxygen	21
Argon and other rare gases	0.9
Carbon dioxide	0.04
Ozone	0.000007

Figure 1**0 4****. 1**

Describe and explain the pattern shown in Figure 1.

[4 marks]

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0 4 . 2

Explain why the atmosphere is described as being in a state of **dynamic equilibrium**.

[2 marks]

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0 4 . 3

Give **two** ways in which the atmosphere supports life other than by providing gases for biological molecules.

[2 marks]

1

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2

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0 4 . 4

Name the two layers of the atmosphere most affected by human activities.

[2 marks]

1

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2

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10

0 5

Table 1 shows the approximate energy density of some energy resources.

Table 1

Energy resource	Energy density / MJ kg ⁻¹
Uranium (natural, in fission reactor)	500 000
Natural gas	55
Petrol	46
Coal	28
Dry wood	16
Lithium-ion battery (stored electricity)	0.7

05 . 1

Explain what is meant by **energy density** and why it is an important feature of an energy resource.

[3 marks]

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05 . 2

Use Table 1 to calculate how many tonnes of coal contain the same energy as 1 kg of uranium.

[2 marks]

Answer

tonnes

05 . 3

Explain why low energy density does not prevent solar and wind power from being useful.

[2 marks]

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05 . 4

Explain the difference between an **intermittent** resource and an **unpredictable** resource, using named examples.

[2 marks]

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05 . 5

Explain why natural gas must be liquefied for transport by ship.

[1 mark]

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10

06 . 1

Describe how soil texture is determined using a stack of sieves.

[3 marks]

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06 . 2

Explain why a soil auger gives a more representative sample than a trowel.

[2 marks]

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06 . 3

Explain why the organic matter content will be overestimated if a soil sample is heated above 550 °C.

[2 marks]

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06 . 4

Describe how a turbidimeter measures water turbidity and explain how quantitative results are obtained.

[3 marks]

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10

07 . 1

Give **three** advantages of incinerating domestic waste.

[3 marks]

1

2

3

07 . 2

Explain how each of the following features of a modern incinerator reduces pollution.

[4 marks]

High combustion temperature with rapid cooling of gases

Water spray scrubber

Removal of wet / compostable waste

Removal of recyclable waste

07 . 3

Explain why the ash from waste incinerators needs careful disposal.

[2 marks]

07 . 4

State **one** reason why incineration is expensive.

[1 mark]

10

08 . 1

Explain the difference between a **mutagen**, a **carcinogen** and a **teratogen**.

[3 marks]

08 . 2

Explain why a birth abnormality caused by a teratogen cannot be inherited by future generations, whereas one caused by a mutagen may be.

[2 marks]

08 . 3

Distinguish between **somatic** and **gonadic** effects of a mutagen.

[2 marks]

08 . 4

Explain what is meant by **synergism** and give an example.

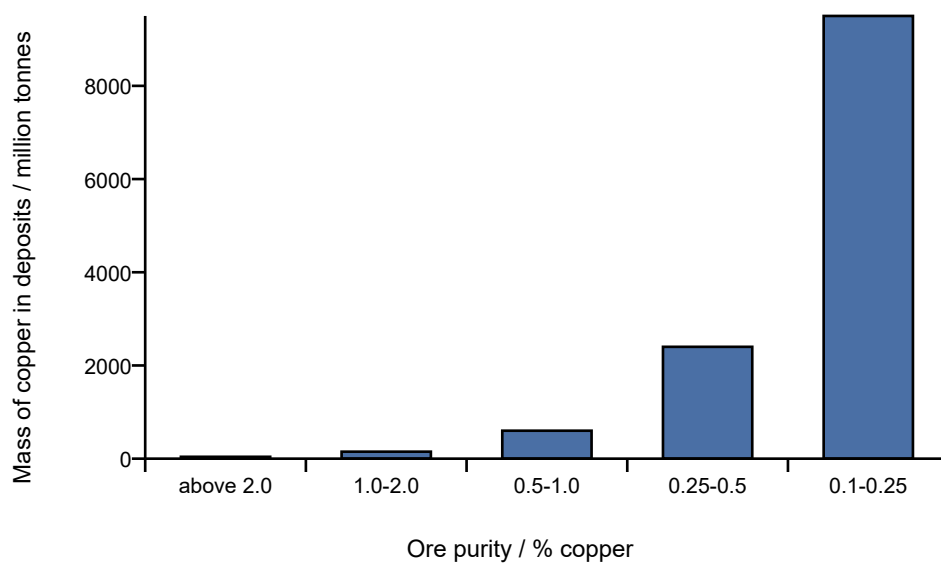
[3 marks]

10

09

Figure 1 shows the mass of copper contained in the Earth's crust at different ore purities, illustrating Lasky's principle.

Figure 1



09

. 1

State Lasky's principle.

[1 mark]

09

. 2

Use Figure 1 to calculate how many times more copper is contained in deposits of 0.25-0.5% purity than in deposits above 2.0%.

[1 mark]

Answer times

0 9 . 3 Explain the difference between the **reserves** and the **resource** of a mineral.
[3 marks]

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0 9 . 4 Explain, using Lasky's principle, why the ability to exploit slightly lower grade ores greatly increases reserves.
[2 marks]

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0 9 . 5 Explain why the main problem for future mineral supplies is not the quantity that exists in the crust.
[3 marks]

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10

10 . 1

Explain why the development of energy storage technologies is essential if renewable energy resources are to provide most of a country's electricity.

[9 marks]

Extra space

10 . 2 State what is meant by **peak shaving**.

[1 mark]

10

Write an essay on one of the following topics.

1	1	.	1
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Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.

[25 marks]

OR

1 1 . 2

Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS